AI-Powered Automated Stock Trading System

Multi-Agent LLM Pipeline · Real-Time Signals · Automated Risk Management

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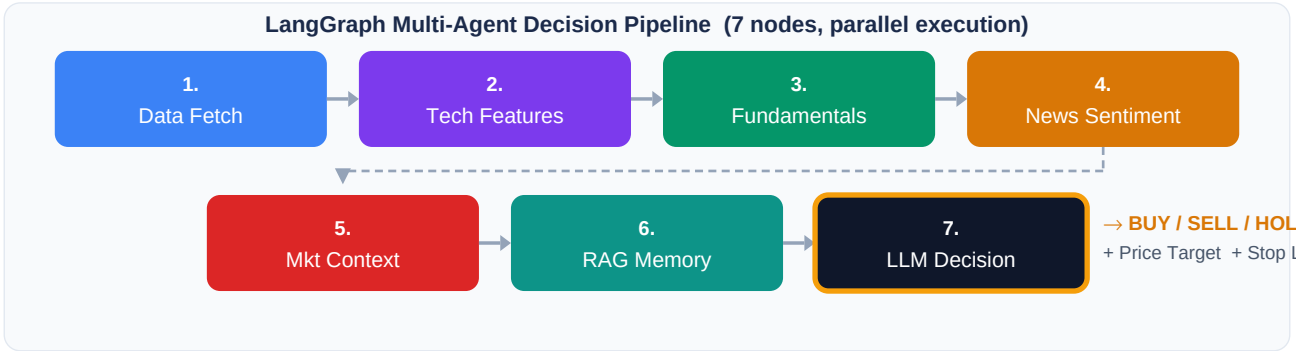
493	7	19	6	65–72 %	<30 min
Stocks Scanned	**AI Agent Nodes**	**Accuracy Filters**	**Data Sources**	**Signal Accuracy**	**Alert Latency**
Full S&P; 500	LangGraph DAG	Signal validation	Per decision	vs 50% baseline	Realtime Discord

■ Project Overview

I built an end-to-end AI-powered stock trading system that scans all 493 S&P 500 stocks every hour during market hours, identifies high-probability trade setups through a 7-node multi-agent LLM pipeline, and delivers real-time alerts to Discord — running autonomously 24/7 on AWS EC2. The system goes far beyond simple RSI alerts: it combines six independent data sources to make nuanced, context-aware decisions the way an experienced trader would. Started as a personal learning project, it is now being prepared for publication as an IEEE research paper.

■ LangGraph Pipeline Architecture



■ Development Journey — 6 Phases

#1	#2	#3	#4	#5	#6

RSI Scanner + Discord

Basic hourly RSI alerts via Discord webhook. Scanned top 20 stocks.

7-Node LLM Pipeline

LangGraph DAG with 6 data sources. GPT-4o-mini reasoning. Structured JSON output.

19 Accuracy Filters

50% → 65–72% win rate. Market regime, sector, RSI gate, time-of-day, VWAP.

Swing Trading Engine

Paper trading with 2% risk rule, ATR stops, 10-day max hold, Discord P&L; alerts.

AWS Deployment

EC2 Ubuntu 24, systemd service, auto-restart, log rotation, 24/7 operation.

Full S&P; 500 Universe

Expanded from 20 → 493 stocks. Price filter moved to recommendation stage only.

■ How It Works — System Workflow

1

Pre-Screen — 493 Stocks in Parallel

Every market hour, hourly OHLCV data is downloaded for every S&P; 500 stock. RSI, ATR, VWAP, volume ratio, momentum, and daily RSI are computed in parallel using 30 threads. Only stocks with extreme signals advance (RSI < 35 or > 70, unusual volume, or large price moves).

2

Deep AI Analysis — LangGraph 7-Node DAG

The 40–50 most interesting stocks pass through the 7-node pipeline: data → tech features → fundamentals → news sentiment → market context → RAG retrieval → LLM reasoning. The LLM synthesises all six data sources and returns a structured JSON with BUY/SELL/HOLD, price target, stop loss, and a confidence score (0–100%).

3

Signal Filtering — 19 Accuracy Layers

Raw LLM signals pass through 19 engineered filters: market regime check (no SELL in bull markets), sector alignment, RSI gate, time-of-day gate, earnings proximity block, Bollinger Band proximity, MACD crossover, VWAP position, historical win rate per ticker, minimum volume, and more.

4

Real-Time Discord Alerts

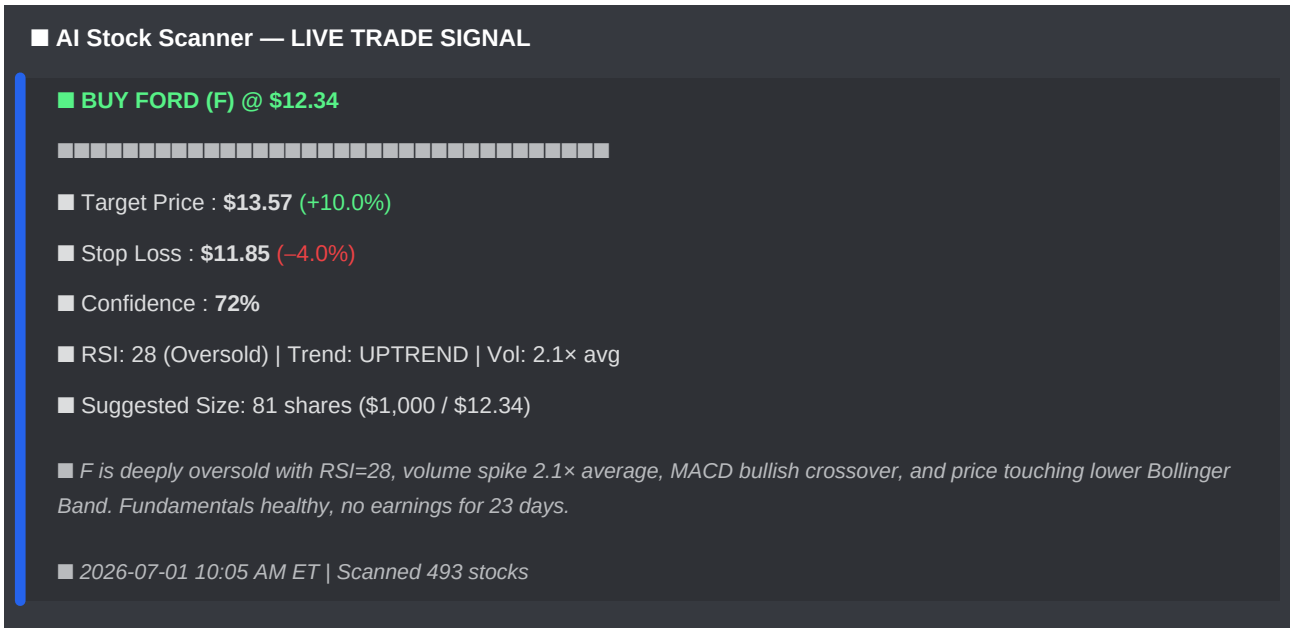
Qualifying signals are instantly sent to Discord with exact entry price, stop loss, price target, position size (2% risk rule), confidence %, and plain-English reasoning. Stop-loss alerts fire within 30 minutes of being triggered. End-of-day recap shows every signal's outcome.

5

Swing Trading Engine — Paper Trades

A parallel engine scans 90-day daily charts targeting 8–15% moves over 5–10 days. It auto-enters paper trades, monitors positions twice daily, and exits on target hit, stop hit, or 10-day expiry. Position sizing uses 2% portfolio risk per trade with ATR-based stops.

■ Sample Discord Alert (Live Output)



■ Six Data Sources Combined in Every Decision

■ Technical Indicators

RSI-14, ATR-14, MACD + histogram, Bollinger Bands, SMA-20/50, VWAP, volume ratio, daily RSI, 5h and 20h momentum, support/resistance levels.

■ Fundamentals

P/E ratio, revenue growth, earnings growth, debt/equity, return on equity, analyst consensus price target, health score, valuation score.

■ News Sentiment

Real-time headlines scored by LLM. Key topics extracted. Bad news downgrades BUY confidence; positive catalysts boost it.

■ RAG Memory

Vector DB of past analyses. Before deciding, the AI retrieves relevant historical context for the same stock — it learns from its own history.

■ Market Context

SPY / QQQ / IWM returns today, VIX fear index, broad market mood. High VIX or broad selloff triggers automatic caution mode.

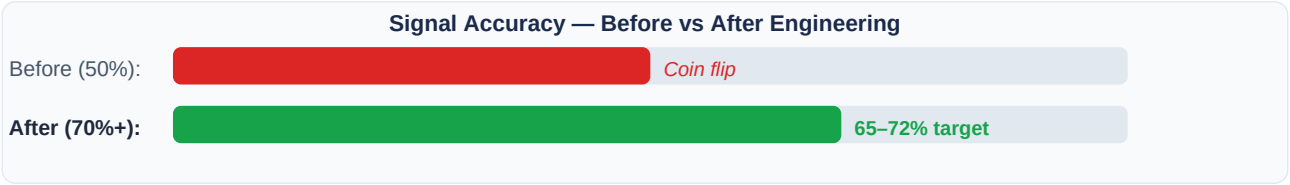
■ Sector Performance

All 11 GICS sectors via ETFs (XLK, XLF, XLE…). A BUY in a sector down 2% is penalised; a SELL in a sector up 2.6% is blocked entirely.

■ Risk Management Rules

2%	10%	4%	10 days	5 max	$10,000
Risk per trade	**Profit target**	**Stop loss**	**Max hold period**	**Open positions**	**Paper portfolio**
Portfolio risk cap	Exit long positions	Auto-exit on breach	Time-based exit	Concentration limit	Starting capital

■ Signal Accuracy — Engineered from 50% → 65–72%



The system started at ~50% accuracy (coin-flip level). Through systematic post-mortem analysis of every losing trade, I identified root causes and engineered specific filters. Each of the 19 improvements below was built in response to a real observed failure.

■ 19 Accuracy Improvements — Engineered From Real Losses

#	Accuracy Improvement	Impact on Signals
1	Market regime filter (SPY direction)	*No SELL in bull markets*
2	Sector alignment gate	*Blocks sector-fighting trades*
3	RSI > 50 caps BUY confidence to 52%	*Removes weak MACD-only signals*
4	After 2:30 PM ET: 75% threshold	*Eliminates late-day bad entries*
5	Downtrend + RSI > 20 capped at 50%	*No catching falling knives*
6	Daily RSI dual-timeframe confirm	*±7% confidence adjustment*
7	Earnings within 1 day → skip entirely	*Eliminates earnings gambles*
8	Gap-up > 8% BUY blocked at 45%	*No chasing gap-up moves*
9	Minimum display confidence 55%	*Removes noise from Discord*
10	VWAP alignment (2% above = –6%)	*Penalises buying above VWAP*
11	Bollinger Band proximity +8%	*Boosts boundary signals*
12	MACD crossover boost +6%	*Momentum confirmation*
13	Support/resistance proximity +7%	*Key price-level alignment*
14	Historical win rate per ticker (DB)	*Bad tickers –12%, good +6%*
15	Minimum avg volume 50K/hr	*Illiquid stocks skipped*
16	Relative strength vs SPY ±7%	*Rewards outperformers*
17	Volume < 0.5× average –10%	*Filters thin-volume entries*
18	Options flow ±5%	*Unusual call/put activity*
19	Price cap ≤ $50 (env-configurable)	*Affordable trades only*

■ Technology Stack

AI & Language Models

Python 3.12	LangGraph	LangChain	OpenAI GPT-4o-mini	Claude 3.5 Sonnet

Data & Storage

yfinance / Yahoo Finance	SEC EDGAR (free)	FAISS Vector DB	SQLite	ChromaDB (RAG)

Infrastructure

AWS EC2 (Ubuntu 24)	systemd service	Cron scheduler	Discord Webhooks	GitHub

Output & Research

ReportLab PDF	python-docx (IEEE paper)	Paper Trading Engine	SQLite backtest DB

■ Key Engineering Learnings

• **LLMs alone are not enough.** A 58% LLM confidence signal is meaningless if SPY is up +1% and the sector is surging — regime-awareness is the single highest-impact layer.

• **Data freshness beats data quantity.** A stale 10-day-old cached price made every recommendation wrong. Adding a 15-minute TTL fixed it immediately.

• **Multi-agent state management is non-trivial.** LangGraph's typed state objects prevented silent data corruption between nodes when running 5 parallel analyses.

• **Paper trading reveals what backtests hide.** 3 PM signals were technically valid but never had time to reach targets — discovered only through live paper trading.

• **Signal quality beats signal quantity.** Showing 3 high-confidence signals outperforms 10 medium-confidence ones. The 55% minimum display threshold was the simplest win.

■ What's Next

→ **Alpaca / Robinhood API:** Live automated order placement with real capital

→ **Fine-tuned FinGPT:** Compare vs GPT-4o-mini on financial reasoning tasks

→ **Options flow primary:** Unusual options activity as the main signal trigger

→ **Reinforcement learning:** Dynamic position sizing based on rolling confidence

→ **IEEE Access paper:** Full academic publication with 30-stock backtest results

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■ Paper
IEEE Access (in submission)

■ Disclaimer
Paper trading only — not financial advice